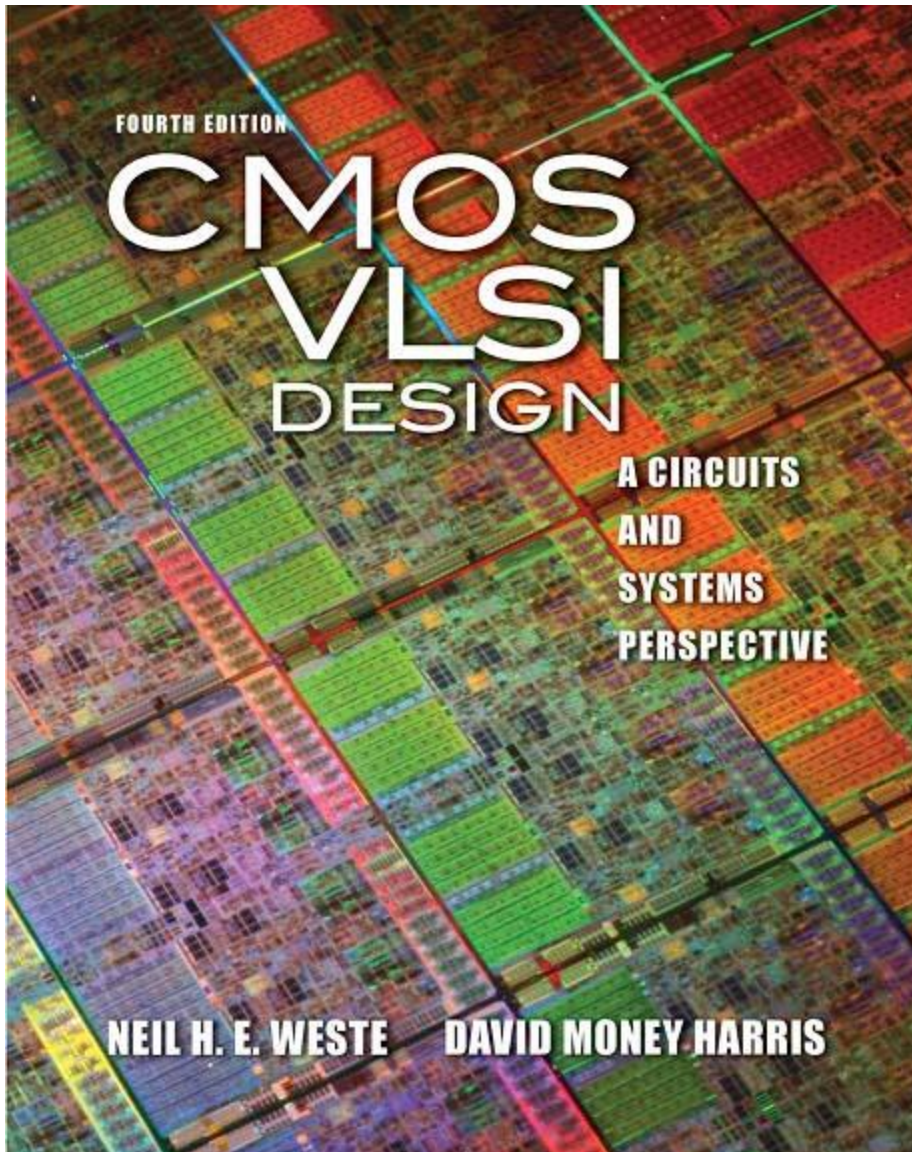
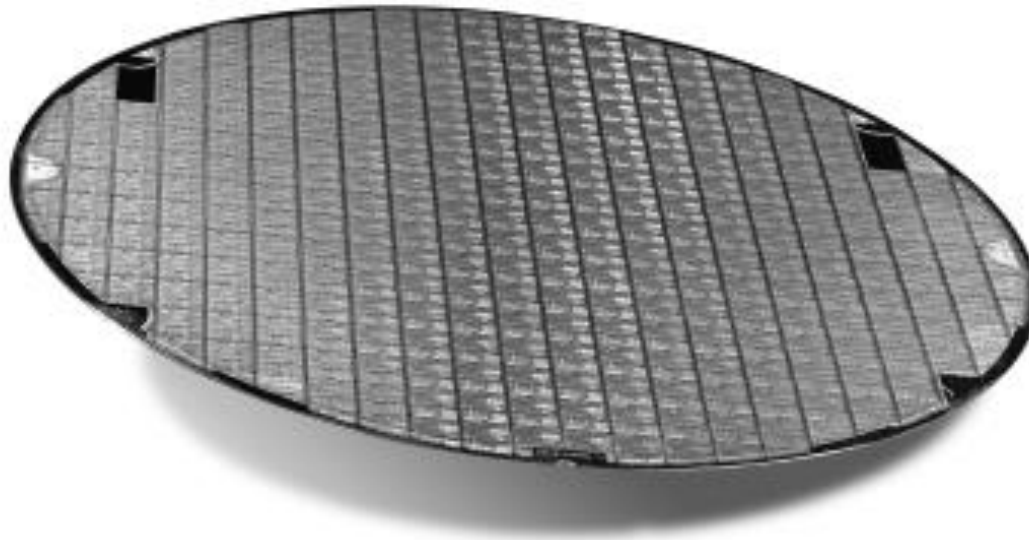




Cost of an Integrated Circuit

Silicon Wafer

- ❑ CMOS transistors are fabricated on silicon wafer (Disk of Silicon)
- ❑ Wafers diameters (75-200-300 mm) and less than 1 mm thick



Total Cost of an IC

- ❑ The total cost of any product can be separated into two components:
 - Recurring expenses(variable cost)
 - Non-recurring expenses (fixed cost)

Fixed Cost

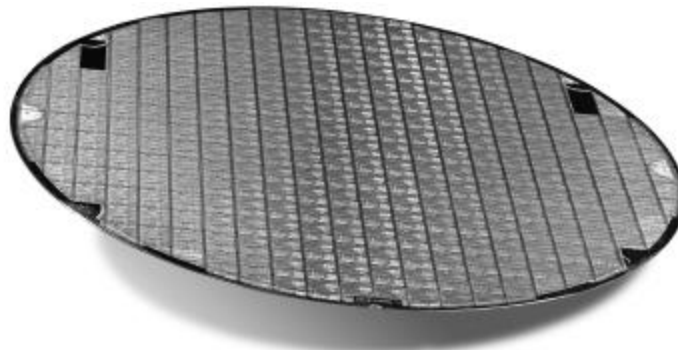
- ❑ **Fixed Cost:** The effort in time and manpower it takes to produce the design. This design cost is strongly influenced by the complexity of the design, the aggressiveness of the specifications, and the productivity of the designer.
- ❑ The indirect costs:
 - company's research and development (R&D)
 - manufacturing equipment
 - marketing
 - sales
 - building infrastructure

Variable Cost

- ❑ **Variable Cost:** This accounts for the cost that is directly attributable to a manufactured product, and is hence proportional to the product volume.
- ❑ Variable costs include:
 - costs of the parts used in the product
 - assembly costs
 - testing costs

Cost of an IC

- ❑ A 200 mm diameter wafer costs \$1000 to process. It is assumed to contain 1 defect/cm² that will impact the product yield. A CMOS chip being fabricated on this wafer occupies an area of 10 mm x 10 mm, or alternately, it may be constructed with 4 identically sized smaller chips of size 6 mm x 6 mm. The package cost for the 10 mm x 10 mm chip is \$ 15.00, whereas the package cost of the smaller chip is \$2.00. The testing cost for each die (prior to packaging) is \$1.50, for all chips tested – both good and bad. What is the cheapest solution?



Parameters

- ☐ Wafer Diameter = 200 mm = 20 cm
- ☐ Cost of Wafer = \$1000
- ☐ Defects = 1 Defect/cm²

- ☐ Chip Area = 10 mm x 10 mm = 100 mm² = 1 cm²
- ☐ Chip Area = 6 mm x 6 mm = 36 mm² = 0.36 cm²

- ☐ Packaging Cost for 100 mm² Chip = \$15
- ☐ Packaging Cost for 36 mm² Chip = \$2

- ☐ Testing Cost (prior to packaging) for both of Chips' Types = \$1.50

- ☐ Cheapest Solution = ?

Cheapest Solution

- ❑ $\text{Cost/Chip} = \frac{\text{Process Cost/Wafer}}{\text{No. of good Chips/Wafer}} + \text{Testing Cost} + \text{Packaging Cost}$
- ❑ $\text{No. of good Chips/Wafer} = \text{No. of Chip/Wafer} \times \text{Yield}$
- ❑ $\text{No. of Chip/Wafer} = (\text{Area of Wafer/Chip Area}) \times \text{Utilization of Wafer}$
- ❑ Assumption $\rightarrow$ 80 % Utilization of Wafer
- ❑ Yield Model = $e^{-\sqrt{AD}}$
- ❑ Where A is the chip area & D is the Defects/cm²

Home Work

- ❑ A 200 mm diameter wafer costs \$1000 to process. It is assumed to contain 1 defect/cm² that will impact the product yield. A CMOS chip being fabricated on this wafer occupies an area of 10 mm x 10 mm, or alternately, it may be constructed with 4 identically sized smaller chips of size 6 mm x 6 mm. The package cost for the 10 mm x 10 mm chip is \$ 15.00, whereas the package cost of the smaller chip is \$2.00. The testing cost for each die (prior to packaging) is \$1.50, for all chips tested – both good and bad. What is the cheapest solution?